

Indexed Files.

Creating an Indexed File

```
      $ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID.    CreateIndexedFromSeq.
* Creates an indexed file  from a sequential file.

ENVIRONMENT DIVISION.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
    SELECT VideoFile ASSIGN TO "VIDEO.DAT"
        ORGANIZATION IS INDEXED
        ACCESS MODE IS RANDOM
        RECORD KEY IS VideoCode
        ALTERNATE RECORD KEY IS VideoTitle
            WITH DUPLICATES
        FILE STATUS IS VideoStatus.
    SELECT SeqVideoFile ASSIGN TO "INVIDEO.DAT".

DATA DIVISION.
FILE SECTION.
FD VideoFile.
01 VideoRecord.
    02 VideoCode                PIC 9(5).
    02 VideoTitle               PIC X(40).
    02 VideoSupplierCode        PIC 99.
FD SeqVideoFile.
01 SeqVideoRecord.
    88 EndOfFile VALUE HIGH-VALUES.
    02 SeqVideoCode             PIC 9(5).
    02 SeqVideoTitle            PIC X(40).
    02 SeqVideoSupplierCode      PIC 99.

WORKING-STORAGE SECTION.
01    VideoStatus                PIC X(2).
```

Creating an Indexed File

```
PROCEDURE DIVISION.  
Begin.  
  OPEN INPUT SeqVideoFile.  
  OPEN OUTPUT VideoFile.  
  
  READ SeqVideoFile  
    AT END SET EndOfFile TO TRUE  
  END-READ.  
  PERFORM UNTIL EndOfFile  
    WRITE VideoRecord FROM SeqVideoRecord  
      INVALID KEY DISPLAY "VIDEO STATUS :- ", VideoStatus  
    END-WRITE  
    READ SeqVideoFile  
      AT END SET EndOfFile TO TRUE  
    END-READ  
  END-PERFORM.  
  
  CLOSE VideoFile, SeqVideoFile.  
  STOP RUN.
```

Reading an Indexed File - Sequentially.

```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID.    ReadingIndexedFile.
* Sequential reading of an indexed file

ENVIRONMENT DIVISION.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
    SELECT VideoFile ASSIGN TO "VIDEO.DAT"
        ORGANIZATION IS INDEXED
        ACCESS MODE IS DYNAMIC
        RECORD KEY IS VideoCode
        ALTERNATE RECORD KEY IS VideoTitle
            WITH DUPLICATES
        FILE STATUS IS VideoStatus.

DATA DIVISION.
FILE SECTION.
FD VideoFile
01 VideoRecord.
    88 EndOfFile VALUE HIGH-VALUE.
    02 VideoCode          PIC 9(5).
    02 VideoTitle         PIC X(40).
    02 SupplierCode       PIC 99.

WORKING-STORAGE SECTION.
01 VideoStatus            PIC X(2).
01 RequiredSequence      PIC 9.
    88 VideoCodeSequence  VALUE 1.
    88 VideoTitleSequence VALUE 2.
01 PrnVideoRecord.
    02 PrnVideoCode       PIC 9(5).
    02 PrnVideoTitle      PIC BBBBX(40).
    02 PrnSupplierCode    PIC BBBB99.
```

Reading an Indexed File - Sequentially.

```
PROCEDURE DIVISION.  
Begin.  
  OPEN INPUT VideoFile.  
  
  DISPLAY "Enter key : 1=VideoCode, 2=VideoTitle ->"  
    WITH NO ADVANCING.  
  ACCEPT RequiredSequence.  
  
  IF VideoTitleSequence  
    MOVE SPACES TO VideoTitle  
    START VideoFile KEY IS GREATER THAN VideoTitle  
      INVALID KEY  DISPLAY "VIDEO STATUS :- ", VideoStatus  
    END-START  
  END-IF  
  
  READ VideoFile NEXT RECORD  
    AT END SET EndOfFile TO TRUE  
  END-READ.  
  PERFORM UNTIL EndOfFile  
    MOVE VideoCode TO PrnVideoCode  
    MOVE VideoTitle TO PrnVideoTitle  
    MOVE SupplierCode TO PrnSupplierCode  
    DISPLAY PrnVideoRecord  
    READ VideoFile NEXT RECORD  
      AT END SET EndOfFile TO TRUE  
    END-READ  
  END-PERFORM.  
  
  CLOSE VideoFile.  
  STOP RUN.
```

Reading an Indexed File - Sequentially.

```
RUN OF INDEX-EG2.EXE USING VIDEOCODE KEY
Enter key : 1=VideoCode, 2=VideoTitle ->1
00121 FLIGHT OF THE CONDOR, THE      03
00333 PREDATOR                        02
00444 LIVING EARTH, THE              03
01001 COMMANDO                       02
01100 ROBOCOP                        01
02001 LEOPARD HUNTS IN DARKNESS, A  03
02121 DIRTY DANCING                  04
03031 COMPETENT CREW                 05
03032 YACHT MASTER                   05
04041 OPEN OCEAN SAILING             05
04042 PRINCESS BRIDE, THE            06
04444 LIFE ON EARTH                  03
05051 OVERBOARD                      01
06061 HOPE AND GLORY                 07
07071 AMONG THE WILD CHIMPANZEES     03
08081 WHALE NATION                   03
09091 BESTSELLER                     07
10001 WICKED WALTZING                04
11111 TERMINATOR, THE                02
13301 MASSACRE AT MASAI MARA         03
14032 KNOTTY PROBLEMS FOR SAILORS     05
17001 ALIEN                          07
17002 ALIENS                         07
17041 GARFIELD TAKES A HIKE          06
18001 SURVIVING THE STORM            05
19444 PINOCCIO                       02
```

```
RUN OF INDEX-EG2 USING VIDEOTITLE KEY
Enter key : 1=VideoCode, 2=VideoTitle ->2
17001 ALIEN                          07
17002 ALIENS                         07
07071 AMONG THE WILD CHIMPANZEES     03
09091 BESTSELLER                     07
01001 COMMANDO                       02
03031 COMPETENT CREW                 05
02121 DIRTY DANCING                  04
00121 FLIGHT OF THE CONDOR, THE      03
17041 GARFIELD TAKES A HIKE          06
06061 HOPE AND GLORY                 07
14032 KNOTTY PROBLEMS FOR SAILORS     05
02001 LEOPARD HUNTS IN DARKNESS, A  03
04444 LIFE ON EARTH                  03
00444 LIVING EARTH, THE              03
13301 MASSACRE AT MASAI MARA         03
04041 OPEN OCEAN SAILING             05
05051 OVERBOARD                      01
19444 PINOCCIO                       02
00333 PREDATOR                        02
04042 PRINCESS BRIDE, THE            06
01100 ROBOCOP                        01
18001 SURVIVING THE STORM            05
11111 TERMINATOR, THE                02
08081 WHALE NATION                   03
10001 WICKED WALTZING                04
03032 YACHT MASTER                   05
```

Reading an Indexed File - Directly.

IDENTIFICATION DIVISION.

PROGRAM-ID. ReadingIndexedFile.

* Illustrates direct read on an indexed file by any key

ENVIRONMENT DIVISION.

INPUT-OUTPUT SECTION.

FILE-CONTROL.

 SELECT VideoFile ASSIGN TO "VIDEO.DAT"

 ORGANIZATION IS INDEXED

 ACCESS MODE IS DYNAMIC

 RECORD KEY IS VideoCode

 ALTERNATE RECORD KEY IS VideoTitle

 WITH DUPLICATES

 FILE STATUS IS VideoStatus.

DATA DIVISION.

FILE SECTION.

FD VideoFile.

01 VideoRecord.

 02 VideoCode PIC 9(5).

 02 VideoTitle PIC X(40).

 02 SupplierCode PIC 99.

WORKING-STORAGE SECTION.

01 VideoStatus PIC X(2).

 88 RecordFound VALUE "00".

01 RequiredKey PIC 9.

 88 VideoCodeKey VALUE 1.

 88 VideoTitleKey VALUE 2.

01 PrnVideoRecord.

 02 PrnVideoCode PIC 9(5).

 02 PrnVideoTitle PIC BBBBX(40).

 02 PrnSupplierCode PIC BBBB99.

Reading an Indexed File - Directly.

```
PROCEDURE DIVISION.  
Begin.  
  OPEN INPUT VideoFile.  
  DISPLAY "Chose key VideoCode = 1, VideoTitle = 2 -> "  
    WITH NO ADVANCING.  
  ACCEPT RequiredKey.  
  IF VideoCodeKey  
    DISPLAY "Enter Video Code (5 digits) -> " WITH NO ADVANCING  
    ACCEPT VideoCode  
    READ VideoFile  
      KEY IS VideoCode  
      INVALID KEY DISPLAY "VIDEO STATUS :- ", VideoStatus  
    END-READ  
  END-IF  
  IF VideoTitleKey  
    DISPLAY "Enter Video Title (40 chars) -> " WITH NO ADVANCING  
    ACCEPT VideoTitle  
    READ VideoFile  
      KEY IS VideoTitle  
      INVALID KEY DISPLAY "VIDEO STATUS :- ", VideoStatus  
    END-READ  
  END-IF  
  IF RecordFound  
    MOVE VideoCode TO PrnVideoCode  
    MOVE VideoTitle TO PrnVideoTitle  
    MOVE SupplierCode TO PrnSupplierCode  
    DISPLAY PrnVideoRecord  
  END-IF.  
  CLOSE VideoFile.  
  STOP RUN.
```


Reading an Indexed File - Directly.

```
RUN OF INDEX-EG3.EXE USING VIDEOCODE
Chose key VideoCode = 1, VideoTitle = 2 -> 1
Enter Video Code (5 digits) -> 02121
02121      DIRTY DANCING                                04

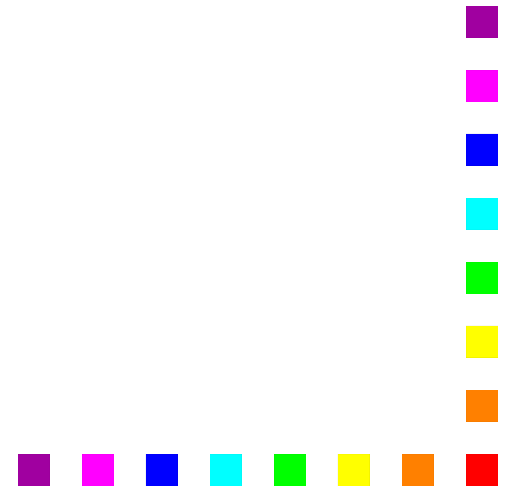
RUN OF INDEX-EG3.EXE USING VIDEOCODE
Chose key VideoCode = 1, VideoTitle = 2 -> 1
Enter Video Code (5 digits) -> 05051
05051      OVERBOARD                                    01

RUN OF INDEX-EG3.EXE USING VIDEOTITLE
Chose key VideoCode = 1, VideoTitle = 2 -> 2
Enter Video Title (40 chars) -> OVERBOARD
05051      OVERBOARD                                    01

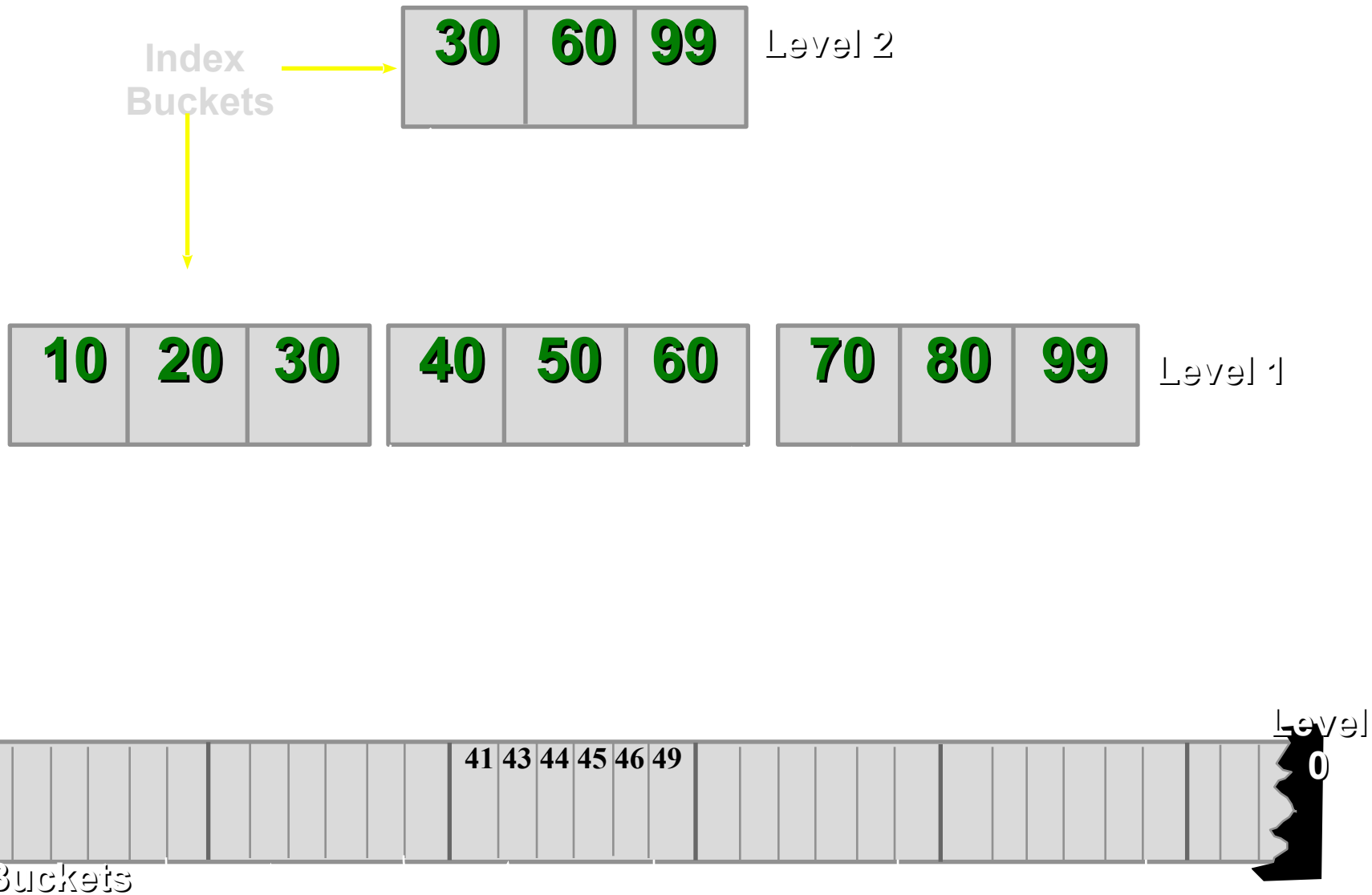
RUN OF INDEX-EG3.EXE USING VIDEOTITLE
Chose key VideoCode = 1, VideoTitle = 2 -> 2
Enter Video Title (40 chars) -> DIRTY DANCING
02121      DIRTY DANCING                                04

RUN OF INDEX-EG3.EXE USING NON EXISTANT VIDEOCODE
Chose key VideoCode = 1, VideoTitle = 2 -> 1
Enter Video Code (5 digits) -> 44444
VIDEO STATUS :- 23
```

Select and Assign for Indexed Files



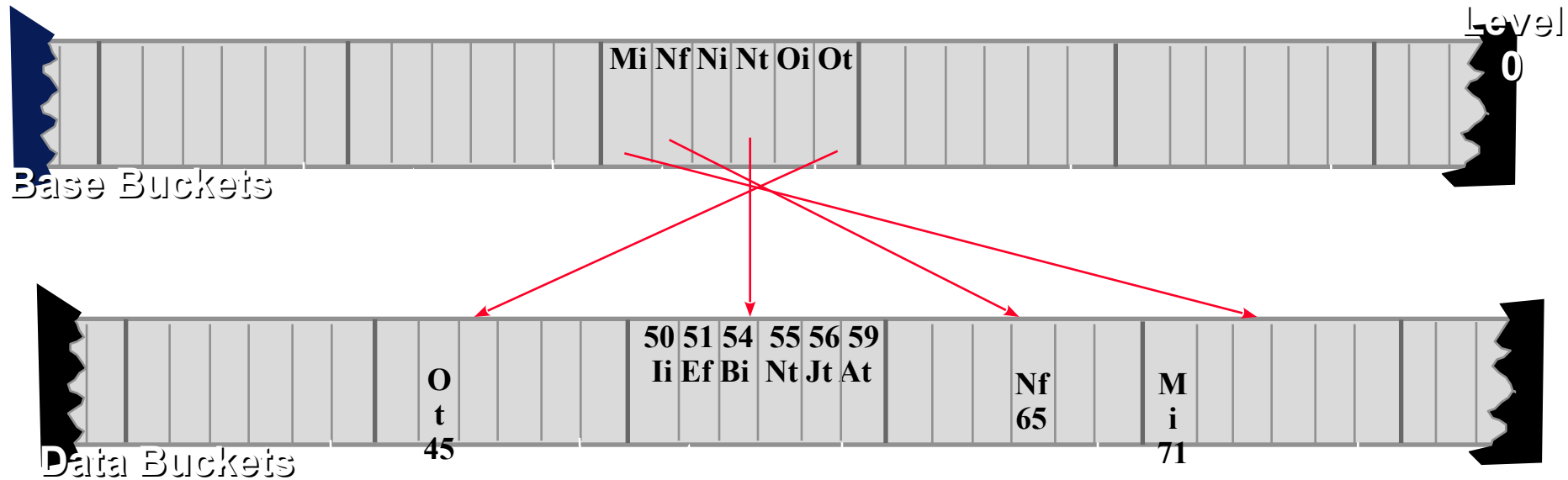
Indexed Files - Primary Key



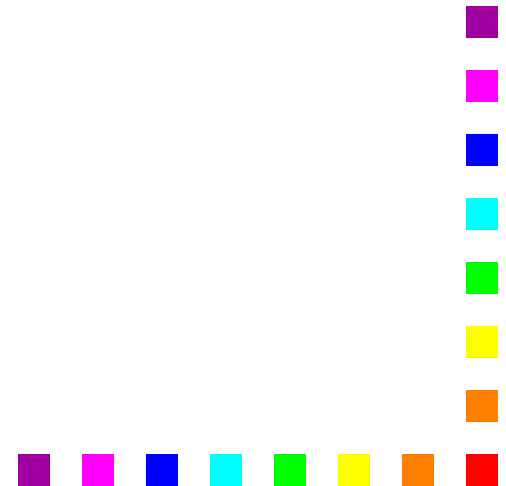
Indexed Files - Alternate Key



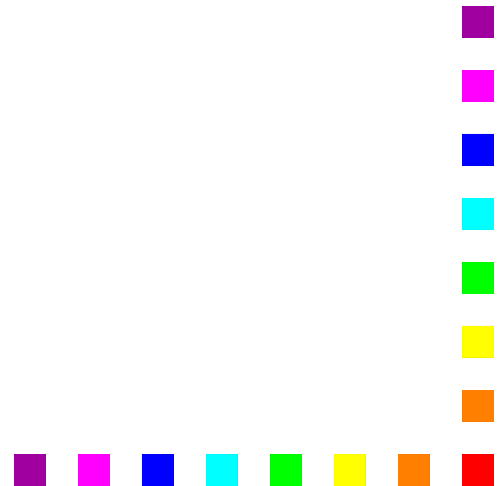
Indexed Files - Alternate Key



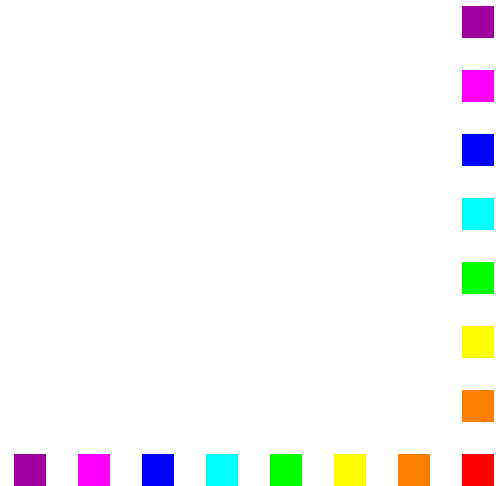
FDs for Indexed Files



Indexed File Verbs - OPEN

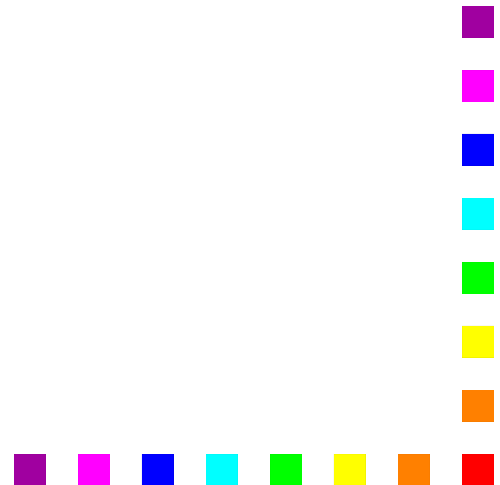


Indexed File Verbs - READ

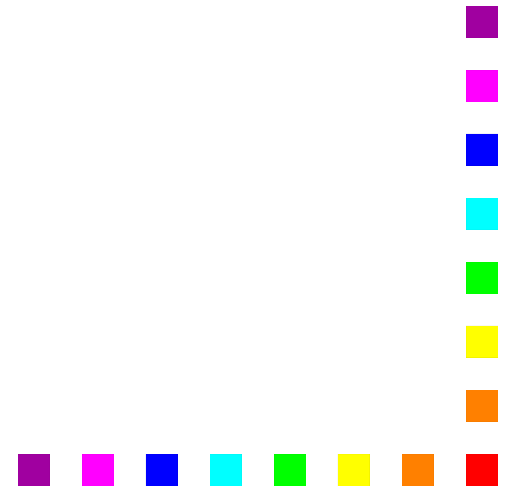


Indexed File Verbs - Write and Rewrite

REWRITE RecName [FROM SourceItem]
[INVALID KEY StatementBlock]
[END-REWRITE]



Indexed File Verbs - DELETE



Indexed File Verbs - START

